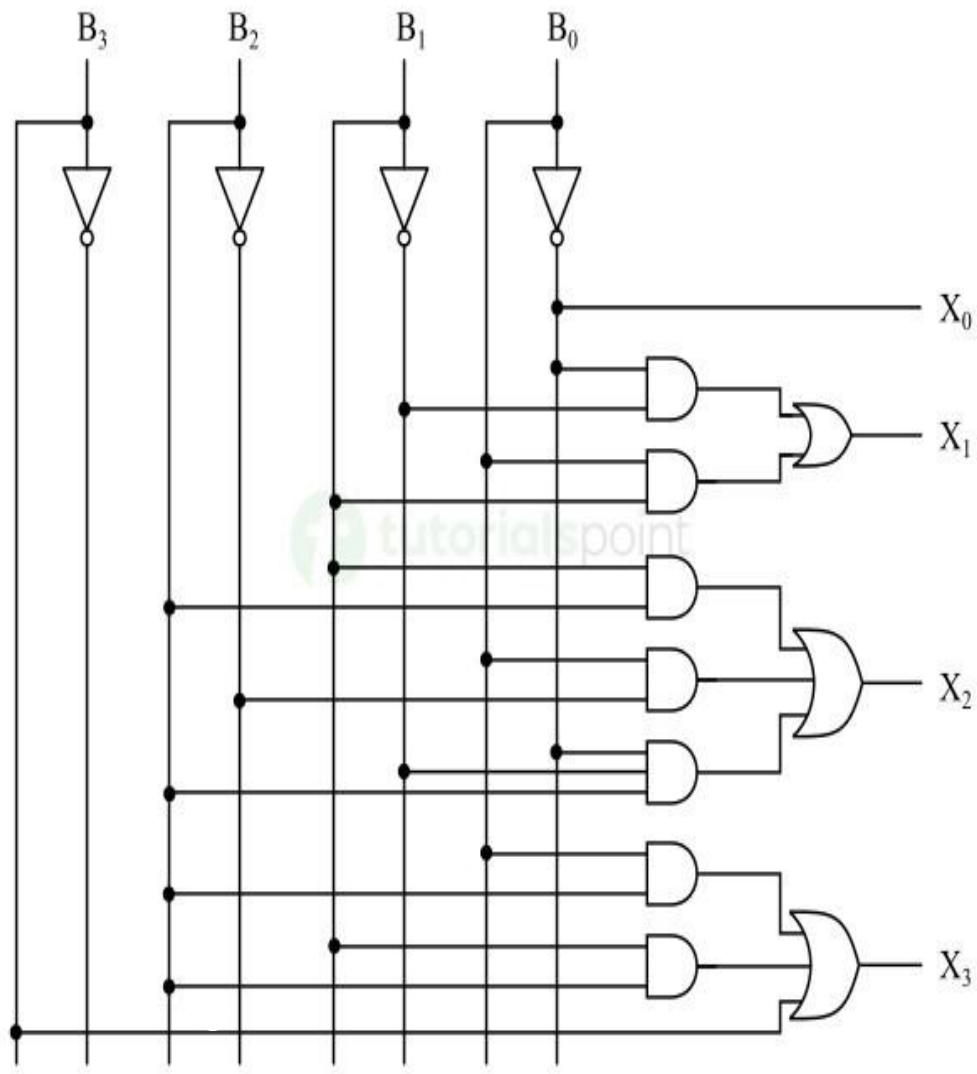


Day 23

BCD to Excess-3 Converter in Verilog

Circuit Diagram :



Verilog Code

BCD to X3 code (Dataflow modeling)

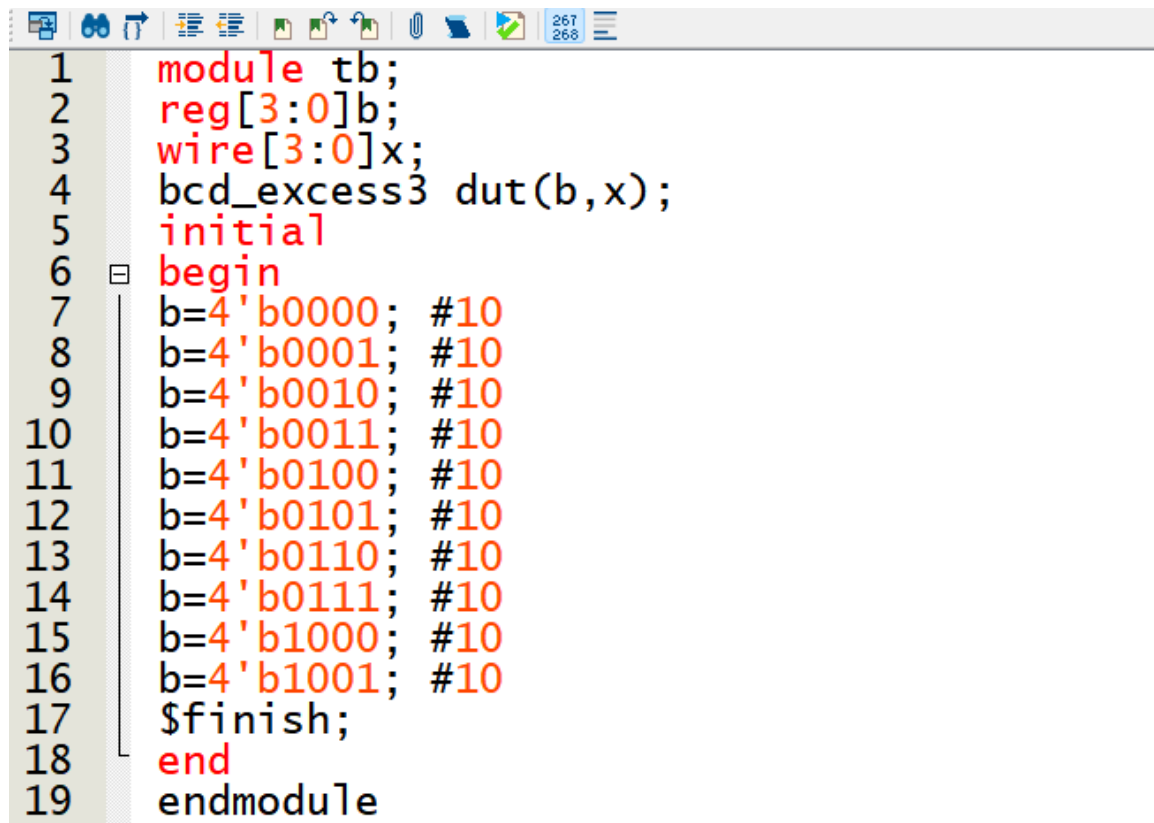
```
1 module bcd_excess3(b,x);
2   input [3:0]b;
3   output [3:0]x;
4   assign x[3]=b[3]|(b[2]&b[1])|(b[2]&b[0]);
5   assign x[2]=(~b[2]&b[0])|(~b[2]&b[1])|(b[2]&~b[1]&~b[0]);
6   assign x[1]=(b[1]&b[0])|(~b[1]&~b[0]);
7   assign x[0]=~b[0];
8   endmodule
9
10
```

BCD to X3 code (Behavioral modeling)

```
1 module bcd_excess3(b,x);
2   input [3:0]b;
3   output reg [3:0]x;
4   always@(*) begin
5     case(b)
6       4'b0000: x = 4'b0011;
7       4'b0001: x = 4'b0100;
8       4'b0010: x = 4'b0101;
9       4'b0011: x = 4'b0110;
10      4'b0100: x = 4'b0111;
11      4'b0101: x = 4'b1000;
12      4'b0110: x = 4'b1001;
13      4'b0111: x = 4'b1010;
14      4'b1000: x = 4'b1011;
15      4'b1001: x = 4'b1100;
16      default: x = 4'bxxxx;
17    endcase
18  end
19 endmodule
20
```

Testbench Code

BCD to X3 Tb code



```
1  module tb;
2  reg[3:0]b;
3  wire[3:0]x;
4  bcd_excess3 dut(b,x);
5  initial
6  begin
7  b=4'b0000; #10
8  b=4'b0001; #10
9  b=4'b0010; #10
10 b=4'b0011; #10
11 b=4'b0100; #10
12 b=4'b0101; #10
13 b=4'b0110; #10
14 b=4'b0111; #10
15 b=4'b1000; #10
16 b=4'b1001; #10
17 $finish;
18 end
19 endmodule
```

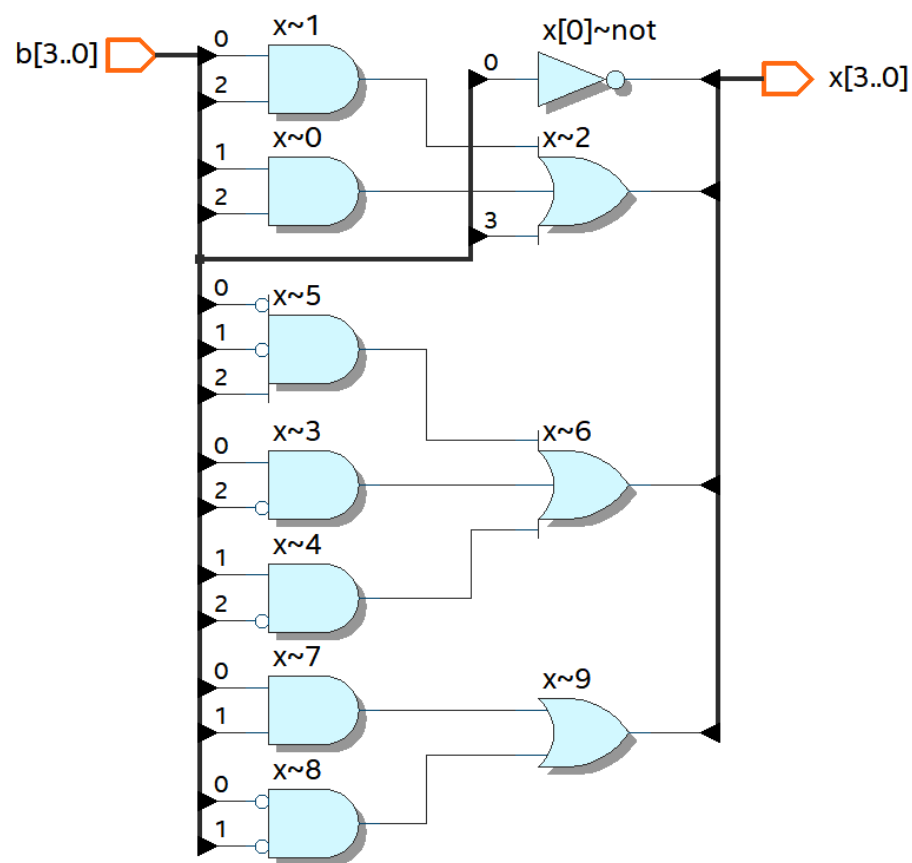
Waveform

Wave - Default

Msgs

<

Schematic



EDA Tools Used

- IntelQuartusPrime
- ModelSim